# Literature Survey

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## Channel Aging in Cell-Free Massive MIMO: Towards Lightweight, Predictive CNN-Based Channel State Estimation for 6G Networks

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*Research Domain: AI-Native Physical Layer Design*

*Research Area: 6G Wireless Communications*

**Refined Problem Statement:**

*"Multi-step forward channel state prediction in distributed cell-free massive MIMO networks using computationally efficient CNNs designed for spatially irregular access point topologies, targeting real-time channel aging mitigation under high user mobility."*

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**Abstract**

This literature survey examines the problem of channel aging in cell-free massive multiple-input multiple-output (MIMO) systems and surveys existing approaches for mitigating it through channel prediction. Beginning with foundational work on massive MIMO channel estimation, the survey traces the evolution from classical model-based predictors — including autoregressive (AR) models and Kalman filters — through early machine learning approaches, to state-of-the-art deep learning architectures employing convolutional neural networks (CNNs), long short-term memory (LSTM) networks, and Transformer models. Special attention is given to the cell-free architecture and the unique challenges it introduces: irregular access point (AP) topology, distributed processing constraints, and the need for computationally efficient edge-deployable models. The survey synthesises findings from over twenty peer-reviewed publications and identifies the critical research gap that the proposed project seeks to address: the absence of lightweight, multi-step predictive CNN architectures specifically designed for the spatially irregular, distributed topology of cell-free massive MIMO systems under high user mobility.

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# 1. Introduction

The demand for wireless connectivity has grown exponentially over the past decade, driven by smartphones, IoT devices, autonomous vehicles, and immersive extended reality applications. Fifth-generation (5G) networks, while a substantial leap forward, are insufficient to meet the projected traffic densities and latency requirements of the 2030 horizon. Sixth-generation (6G) networks are therefore under active research, targeting terabit-per-second peak data rates, sub-millisecond latency, and near-universal seamless coverage [1].

Two of the most promising enabling technologies for 6G are massive multiple-input multiple-output (mMIMO) systems and the cell-free (CF) network architecture. Massive MIMO exploits large antenna arrays — typically 64 to 256 antennas per base station — to serve many users simultaneously through spatial multiplexing and beamforming [2]. The cell-free paradigm extends this concept by distributing a large number of access points (APs) across a wide geographical area, all connected to a centralised processing unit (CPU), collectively serving users without cell boundaries [3, 4].

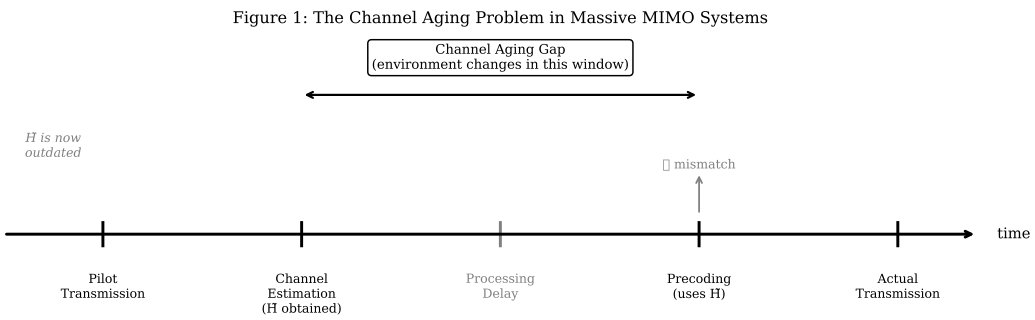


Figure 1: The channel aging problem: the channel state measured at pilot time ($\tilde{\mathbf{H}}$) becomes outdated by the time precoding is applied, leading to beam misalignment and throughput loss.

A fundamental operational requirement for both massive MIMO and cell-free systems is accurate, real-time knowledge of the channel state information (CSI) at the transmitter. CSI describes the complex propagation characteristics between each transmit antenna and each user device, encoding how the signal is attenuated, delayed, and phase-shifted as it travels through the wireless environment. In time-division duplex (TDD) systems, the transmitter estimates the downlink channel by exploiting uplink-downlink reciprocity: the user device transmits known pilot sequences, and the AP measures how these pilots were distorted to reconstruct the channel matrix $\mathbf{H} \in \mathbb{C}^{N \times K}$, where $N$ is the number of antennas and $K$ is the number of subcarriers.

The critical challenge, illustrated in Figure 1, is that the process of pilot transmission, channel estimation, and precoder computation takes several milliseconds. During this

processing delay, the physical environment changes — users move, doors open, vehicles pass — causing the estimated channel $\tilde{\mathbf{H}}$ to become stale relative to the true channel at the time of data transmission. This phenomenon is known as *channel aging*, and it directly degrades beamforming performance, spectral efficiency, and system reliability [5].

Channel aging is particularly pronounced in cell-free systems for two compounding reasons [6]: (1) centralised precoding requires all AP-to-user channel matrices to be gathered at the CPU, introducing additional fronthaul delays on top of the estimation delay; and (2) the distributed architecture means that each user is tracked by multiple independently aging channel links, all of which must remain coherent for joint transmission to succeed.

This survey reviews the body of literature addressing channel aging mitigation through predictive channel state estimation, with the goal of establishing the research context and identifying the gap that motivates the proposed project.

## 2. Massive MIMO: Foundations and Channel Estimation

### 2.1. Overview of Massive MIMO

The concept of massive MIMO was formally popularised by Marzetta (2010) and subsequently analysed in depth by Lu *et al.* [2]. The fundamental insight is that as the number of base station antennas $N$ grows large while the number of served users $K$ remains fixed, inter-user interference averages out and the system approaches a regime of near-orthogonal user channels. This allows simple linear precoders such as maximum ratio transmission (MRT) to approach optimal performance, substantially reducing computational requirements [2].

By the time of 5G standardisation, massive MIMO with 64-element active antenna units had been commercially deployed in sub-6 GHz bands. Looking towards 6G, the concept has been extended to extremely large-scale MIMO (XL-MIMO) with hundreds to thousands of antennas, as surveyed in [1], introducing additional challenges in near-field propagation and spatially non-stationary channel behaviour.

### 2.2. Classical Channel Estimation Methods

Channel estimation in massive MIMO systems relies on pilot sequences: the user device transmits a known signal of length $\tau_p$ symbols, and the receiver computes the channel estimate from the received pilot observations. Two classical estimators are widely used:

#### 2.2.1. Least Squares (LS) Estimation

The LS estimator computes:

$$\hat{\mathbf{H}}^{\mathrm{LS}} = \mathbf{Y}_p \mathbf{\Phi}^{\dagger} \tag{1}$$

where $\mathbf{Y}_p$ is the received pilot matrix and $\mathbf{\Phi}$ is the pilot sequence matrix. The LS estimator is unbiased and computationally efficient but noise-amplifying.

#### 2.2.2. MMSE Estimation

The minimum mean square error (MMSE) estimator incorporates channel statistics to minimise the expected squared error:

$$\hat{\mathbf{H}}^{\mathrm{MMSE}} = \mathbf{R}_H \left( \mathbf{R}_H + \frac{\sigma_n^2}{\rho} \mathbf{I} \right)^{-1} \hat{\mathbf{H}}^{\mathrm{LS}} \tag{2}$$

where $\mathbf{R}_H$ is the channel covariance matrix and $\rho$ is the pilot signal-to-noise ratio (SNR). The MMSE estimator achieves significantly better performance but requires knowledge of $\mathbf{R}_H$, which is often estimated from historical observations [7].

Both methods, however, produce *instantaneous* estimates that do not account for the temporal evolution of the channel. They are estimation methods, not prediction methods, and thus do not address the channel aging problem.

## 3. The Channel Aging Problem

### 3.1. Mathematical Formulation

The time-varying wireless channel is commonly modelled using Jake's isotropic scattering model [8]. Under this model, the channel at time step $t + \delta$ is related to the channel at time $t$ through the normalised Doppler frequency $f_D = f_c v/c$, where $f_c$ is the carrier frequency, $v$ is the user velocity, and $c$ is the speed of light. The temporal correlation coefficient is:

$$\rho(\delta) = J_0(2\pi f_D \delta T_s) \tag{3}$$

where $J_0(\cdot)$ is the zeroth-order Bessel function of the first kind and $T_s$ is the sampling interval. As $\delta$ increases — as more time passes between channel measurement and data transmission — $\rho(\delta)$ decays, quantifying how quickly the channel becomes outdated.

The normalised Doppler frequency $f_n = f_D T_s$ is a key parameter. Table 1 illustrates the channel coherence degradation at different user velocities for a 3.5 GHz 5G-NR carrier.

Table 1: Normalised Doppler frequency and correlation coefficient at 3.5 GHz for different user velocities (5 ms processing delay assumed).

| **Velocity (km/h)** | $f_D$ **(Hz)** | $f_n$ | $\rho(\delta)$ |
|---|---|---|---|
| 3 (pedestrian) | 9.7 | $4.9 \times 10^{-5}$ | $\approx 1.00$ |
| 30 (urban) | 97.2 | $4.9 \times 10^{-4}$ | $\approx 0.98$ |
| 60 (suburban) | 194.4 | $9.7 \times 10^{-4}$ | $\approx 0.94$ |
| 120 (highway) | 388.9 | $1.9 \times 10^{-3}$ | $\approx 0.77$ |
| 300 (high-speed rail) | 972.2 | $4.9 \times 10^{-3}$ | $\approx 0.32$ |

### 3.2. Impact on Spectral Efficiency

The work of Zheng *et al.* [5] provided the first comprehensive closed-form analysis of channel aging in cell-free massive MIMO systems. They derived spectral efficiency (SE) expressions incorporating both imperfect channel estimation and aging under spatially correlated channels, finding that:

- Channel aging causes monotonically decreasing SE as the Doppler spread increases.
- Cell-free systems, while inherently more resilient to aging than small-cell systems due to macrodiversity, still suffer meaningful SE losses at moderate-to-high user velocities.
- Statistical channel cooperation power control (SCCPC) partially mitigates aging effects but its gains erode as aging intensifies.

Similarly, the empirical measurement study by Löschenbrand *et al.* [9] provided real-world validation of channel aging effects in a widely distributed cell-free array with 32 BS antennas and vehicular users moving at 15–60 km/h, confirming analytical predictions and demonstrating that aging is a practical concern, not merely a theoretical artefact.

### 3.3. Cell-Free Specific Aggravating Factors

In a conventional cellular system, the antenna array is co-located, and the spatial structure of the channel is well-defined relative to a single array geometry. In a cell-free system, APs are deployed on lamp posts, building facades, and bus shelters — positions determined by urban infrastructure rather than signal processing optimality. As a result, the inter-AP spatial correlation structure is *irregular*, and channel prediction models designed for regular antenna array geometries do not directly transfer [10].

The work of Freitas *et al.* [6] was among the first to explicitly address channel aging in cell-free MIMO from a prediction perspective, proposing an extended Kalman

filter (EKF)-based directional CSI prediction scheme operating under high Doppler conditions. While effective, the EKF approach requires precise parametric channel modelling, which is difficult to maintain as the propagation environment changes dynamically.

## 4. Cell-Free Massive MIMO: Architecture and Challenges

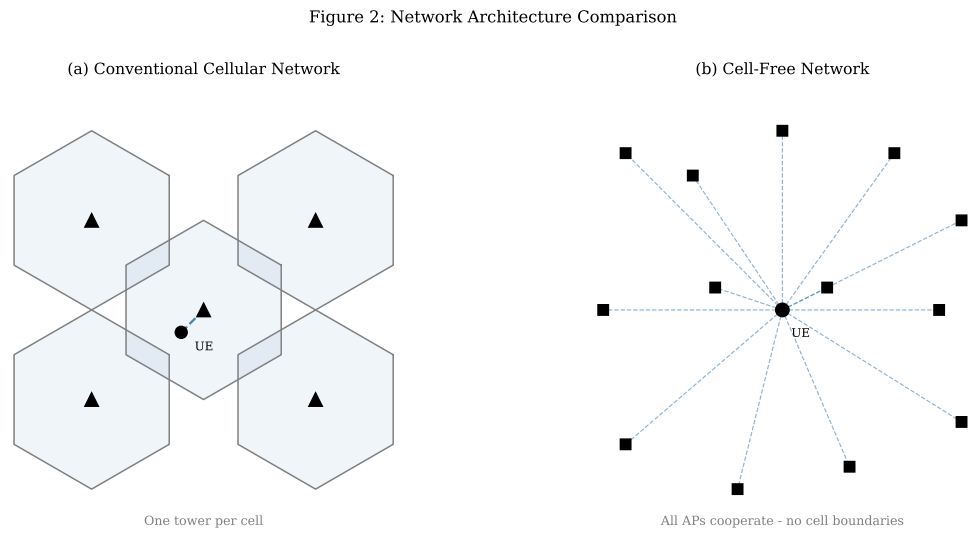


Figure 2: Architecture comparison: (a) conventional cellular network with one base station per cell; (b) cell-free network with distributed APs cooperating to serve all users simultaneously without cell boundaries.

### 4.1. System Model

As illustrated in Figure 2, a cell-free massive MIMO system consists of $L$ single- or multi-antenna APs distributed across a geographical area, connected via fronthaul links to a CPU. All $L$ APs simultaneously serve $K$ user equipment (UE) devices using the same time-frequency resources. The received signal at the CPU from user $k$ on subcarrier $m$ is:

$$\mathbf{y}_{k,m} = \sum_{l=1}^{L} \mathbf{h}_{lk,m}\, x_k + \sum_{j \neq k} \sum_{l=1}^{L} \mathbf{h}_{lj,m}\, x_j + \mathbf{n} \tag{4}$$

where $\mathbf{h}_{lk,m} \in \mathbb{C}^{N_l}$ is the channel vector between AP $l$ and user $k$ on subcarrier $m$, $x_k$ is the transmitted symbol from user $k$, and $\mathbf{n}$ is additive noise.

A comprehensive survey of this architecture, covering fronthaul design, scalability, and 5G/6G integration, is provided in [4]. The seminal work by Ngo *et al.* [3] demonstrated that cell-free systems offer 5-fold and 10-fold improvements over conventional massive MIMO and small-cell systems, respectively, in terms of 95th-percentile per-user throughput. More recent work has extended the architecture to integrate re-

configurable intelligent surfaces (RIS) [11] and to support ultra-reliable low-latency communications (URLLC) for 6G [12].

### 4.2. Channel Estimation in Cell-Free Systems

Channel estimation in cell-free systems follows a TDD pilot-based protocol. In the uplink training phase, each user transmits pilot sequences and each AP estimates the channels from all users simultaneously. The joint channel matrix across all APs is:

$$\mathbf{H} = \begin{bmatrix} \mathbf{H}_1^\top & \mathbf{H}_2^\top & \cdots & \mathbf{H}_L^\top \end{bmatrix}^\top \in \mathbb{C}^{\sum_l N_l \times K} \tag{5}$$

where $\mathbf{H}_l \in \mathbb{C}^{N_l \times K}$ is the channel sub-matrix for AP $l$.

A key challenge is *pilot contamination*: when the number of users $K$ exceeds the pilot sequence length $\tau_p$, pilots must be reused across users, causing inter-user interference in channel estimates. While pilot contamination has been extensively studied in conventional massive MIMO [13], it interacts with channel aging in cell-free systems in ways that are not yet fully characterised in the literature.

## 5. Model-Based Channel Prediction Methods

### 5.1. Autoregressive Models

The autoregressive (AR) model represents the channel as a linear combination of its past values plus a stochastic innovation:

$$\mathbf{h}_t = \sum_{p=1}^{P} \mathbf{A}_p \, \mathbf{h}_{t-p} + \mathbf{e}_t \tag{6}$$

where $\mathbf{A}_p$ are the AR coefficient matrices of order $P$ and $\mathbf{e}_t$ is a zero-mean white noise innovation term. The AR parameters are estimated from observed channel data using the Yule-Walker equations or least squares [8].

AR models provide a computationally efficient prediction baseline. However, their performance degrades rapidly under high mobility because the channel correlation structure changes faster than the AR parameters can be updated. Furthermore, they assume stationarity over the estimation window, which fails in environments with abrupt changes in the scattering geometry.

### 5.2. Kalman Filter-Based Prediction

The Kalman filter (KF) is the optimal linear predictor for linear Gaussian dynamical systems. By casting the AR model in state-space form, the KF recursively updates

the channel prediction as new pilot observations arrive [14]:

$$\hat{\mathbf{h}}_{t|t-1} = \mathbf{F}\,\hat{\mathbf{h}}_{t-1|t-1} \tag{7}$$

$$\hat{\mathbf{h}}_{t|t} = \hat{\mathbf{h}}_{t|t-1} + \mathbf{K}_t\Big(\mathbf{h}_t^{\text{obs}} - \hat{\mathbf{h}}_{t|t-1}\Big) \tag{8}$$

where $\mathbf{F}$ is the state transition matrix derived from the AR coefficients, and $\mathbf{K}_t$ is the Kalman gain.

The work of Kashyap *et al.* [14] established the Kalman filter as an effective channel predictor for TDD massive MIMO, showing that ARMA-based predictors can substantially increase pilot spacing while maintaining acceptable spectral efficiency. The vector Kalman filter (VKF) of Kim *et al.* [8] extended this to the multi-antenna case and provided a comparative benchmark against machine learning approaches.

The EKF-based predictor of Freitas *et al.* [6] represented the first dedicated Kalman-based predictor for cell-free MIMO. However, Kalman-based methods share a fundamental limitation: they require channel model parameters (AR coefficients, noise covariance) to be estimated and updated in real time, creating a computational overhead that scales with the number of APs and users — a significant scaling problem in cell-free systems with $L = 50$–$200$ APs.

### 5.3. Wiener Filter and ARMA Predictors

Wiener filtering provides the minimum mean-square-error linear predictor of a stationary stochastic process. While theoretically optimal for stationary channels, its implementation requires full knowledge of the channel's power spectral density, which is environment-dependent and must be estimated adaptively. ARMA-based predictors approximate the Wiener filter by fitting a higher-order model to the observed temporal correlation [14]. These methods show good performance in low-to-medium mobility scenarios but degrade at high velocities where the stationarity assumption breaks down.

## 6. Deep Learning for Channel Estimation and Feedback

### 6.1. CsiNet: The Foundational Deep Learning CSI Architecture

The breakthrough paper by Wen *et al.* [15] introduced CsiNet, the first deep learning architecture for CSI compression and feedback in FDD massive MIMO. CsiNet uses a CNN-based autoencoder: the encoder (at the user side) compresses the channel matrix into a compact codeword, and the decoder (at the base station) reconstructs the full CSI from the codeword.

CsiNet demonstrated that CNNs can learn efficient representations of the spatial structure of CSI matrices, significantly outperforming compressed sensing-based methods at high compression ratios. Its success established the paradigm of treating the channel matrix as an image-like 2D structure amenable to convolutional processing — a paradigm directly applicable to channel prediction.

Building on CsiNet, Wang *et al.* [16] introduced CsiNet-LSTM, which augments the CsiNet architecture with LSTM cells to exploit temporal correlations across consecutive channel snapshots. CsiNet-LSTM significantly improves reconstruction quality in time-varying channels by learning both spatial structure (via CNN) and temporal dynamics (via LSTM), establishing the hybrid spatial-temporal architecture as a natural design paradigm for channel-related deep learning tasks.

### 6.2. Deep CNN for Wideband Channel Estimation

Dong *et al.* proposed a deep CNN specifically for wideband millimetre wave (mmWave) massive MIMO channel estimation, exploiting the frequency correlation structure across OFDM subcarriers. Their architecture processes the channel matrix across antenna and frequency dimensions simultaneously, extracting both spatial and spectral features. This work demonstrated that the multi-dimensional structure of the channel matrix ($N_{\text{ant}} \times N_{\text{sub}} \times N_{\text{time}}$) is naturally suited to 3D convolutional processing.

A lightweight variant was later proposed by Tian *et al.* [17], who combined CNN with LSTM in a regularisation-enhanced hybrid architecture for massive MIMO channel estimation, demonstrating improved robustness to noise while maintaining lower computational complexity than Transformer-based competitors.

### 6.3. Joint Channel Estimation and Feedback

Guo *et al.* [13] proposed a unified deep learning framework that jointly performs channel estimation, compression, and feedback in a single end-to-end trained network, avoiding the suboptimality introduced by separately optimising each module. Their framework achieves better spectral efficiency than two-stage approaches while reducing overall latency — an important property for channel-aging-aware systems.

## 7. Machine Learning for Channel Prediction

### 7.1. Early ML-Based Prediction: MLP and Basic CNN

The systematic comparison study of Kim *et al.* [8] was the first to compare Kalman filter and machine learning approaches for massive MIMO channel prediction on realistic 3GPP spatial channel model (SCM) channels. Their results established several

important benchmarks:

- A multilayer perceptron (MLP) with LMMSE pre-processing approaches the performance of the VKF at low complexity.
- A CNN-AR architecture (CNN with autoregressive pre-processing) outperforms CNN-RNN in low-to-medium mobility scenarios.
- All ML-based architectures degrade at high normalised Doppler ($f_n > 0.2$), indicating the need for more powerful models.

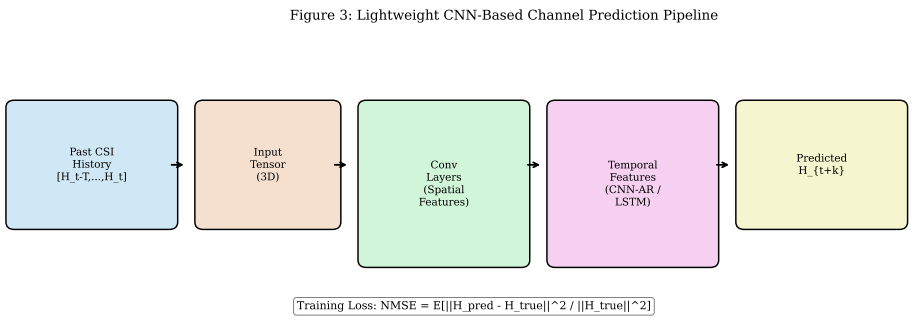


Figure 3: Proposed CNN-based channel prediction pipeline: a sequence of past CSI matrices is fed into the CNN, which outputs the predicted future channel state $\hat{\mathbf{H}}_{t+k}$ for use in precoding.

### 7.2. CNN-Based Real-Time Prediction for High-Speed Scenarios

Targeting high-speed railway and vehicular scenarios, a novel real-time CNN channel prediction algorithm was proposed in [18], which uses reference signal-based online training to continuously adapt to non-stationary channel variations. The algorithm updates its model using only the most recent pilot observations, enabling tracking of fast time-varying channels in LTE and 5G OFDM systems without requiring pre-collected training datasets.

This work demonstrated that CNNs can learn the temporal feature structure of channels directly from reference signals, making them suitable for online deployment — a property important for cell-free systems where channel conditions vary continuously across the distributed AP topology.

### 7.3. Wideband Channel Prediction and Frequency Correlation

The machine learning channel prediction study of Ko *et al.* [19] explored wideband massive MIMO channel prediction, proposing domain transformation techniques to reduce redundancy across subcarrier channels. Their joint learning approach trains a single predictor across all subcarriers, exploiting frequency correlation to reduce data requirements and improve generalisation. This work also identified ML-based

channel prediction for cell-free massive MIMO as an explicit future research direction, noting that the combination of wideband operation, distributed AP topology, and high mobility creates a prediction problem qualitatively different from single-cell massive MIMO.

### 7.4. Hybrid CNN-LSTM Architectures

The work of Nguyen *et al.* [20] proposed a CNN-LSTM model for channel estimation in RIS-NOMA assisted 6G networks, demonstrating that combining CNN (for spatial feature extraction) with LSTM (for temporal sequence modelling) yields superior performance compared to either component alone. The model was shown to be robust to varying system parameters including transmit SNR, power allocation, and the number of RIS elements — a form of generalisation desirable for cell-free systems where channel statistics vary across AP locations.

Building on similar principles, the uplink OFDM channel predictor of De Filippo *et al.* [10] extended the CNN-LSTM encoder-decoder framework with skip connections for 6G non-terrestrial networks, predicting the channel frequency response matrix for entire OFDM slots without pilots. While targeted at satellite scenarios, this work directly demonstrated the viability of CNN-LSTM prediction for distributed, multi-AP scenarios.

### 7.5. Mobile Cell-Free MIMO: Prediction Challenges

The mobile cell-free massive MIMO challenges survey by Zheng *et al.* [21] reviewed prediction methods in the context of distributed systems. They noted that both Kalman filter and ML-based predictors degrade in high-mobility scenarios beyond prediction horizons of approximately 0.1–0.3 wavelengths. They proposed the concept of the "predictor antenna" — a physically forward-deployed antenna that observes the channel the mobile unit will encounter in the near future — as an alternative paradigm, but acknowledged that this approach requires hardware modifications to the AP infrastructure, limiting its practical appeal.

## 8. Transformer and Attention-Based Approaches

### 8.1. Transformer-Based Channel Prediction

The Transformer-based channel predictor of Jiang *et al.* [22] represented a significant advance, proposing a parallel synchronous prediction framework that simultaneously predicts multiple future channel frames rather than sequentially predicting one step at a time. Their hybrid attention mechanism combines self-attention across the time

dimension with positional encoding adapted to channel statistics, achieving prediction NMSE below $-15$ dB for user speeds up to 60 km/h on 3GPP CDL-B and UMa channel models.

However, the Transformer architecture introduced several practical limitations in the cell-free context:

1. Computational complexity is $\mathcal{O}(L^2)$ in sequence length, which becomes prohibitive for long prediction horizons.
2. The model size (millions of parameters) is incompatible with deployment on resource-constrained edge hardware at distributed APs.
3. The model requires extensive matched-condition training data and exhibits reduced generalisation across different channel environments.

A subsequent study on space-time-frequency correlation exploitation for cell-free MIMO [19] reinforced this conclusion, noting that FreqConv and SpaceConv based Transformer encoders — while achieving excellent accuracy in centralised systems — face practical difficulty in extracting spatial correlations under the irregular AP deployment geometry characteristic of cell-free networks.

### 8.2. Attention Mechanisms for Selective Estimation

Gao *et al.* [23] proposed an attention-aided deep learning framework for massive MIMO channel estimation that uses a "divide-and-conquer" strategy: an attention mechanism first identifies the most informative angular regions of the channel, then focuses estimation resources on those regions. This attention-guided approach substantially improves estimation accuracy for channels with narrow angular spread at modest complexity overhead. The result suggests that selective attention mechanisms can complement lightweight CNN architectures without prohibitive computational cost — an idea potentially applicable in the proposed cell-free prediction design.

## 9. AI-Native Physical Layer Design for 6G

The concept of AI-native physical layer design represents a paradigm shift in wireless communication: rather than applying machine learning as a post-hoc optimisation on conventionally designed systems, the physical layer is designed from the ground up to be learned and adapted by AI models. This vision, articulated in the 6G research community and partly incorporated into 3GPP Release 18 standardisation activities for 5G-Advanced, encompasses AI-based channel estimation, AI-based channel prediction, AI-based beamforming, and AI-based modulation and coding [24].

In this context, lightweight CNN-based channel prediction is not merely a techni-

cal optimisation problem but a fundamental building block of the AI-native physical layer. The proposed research contributes to this vision by demonstrating that computationally efficient deep learning can replace or substantially outperform classical model-based predictors even in the most demanding deployment scenario: the spatially distributed, high-mobility cell-free network.

The challenges and opportunities of machine learning for future wireless communications, including channel prediction for cell-free systems, are comprehensively reviewed in [25]. That survey explicitly lists wideband CSI prediction for cell-free massive MIMO exploiting space-time-frequency correlations as an open research problem, directly validating the novelty and timeliness of the proposed project.

The broader 6G AI and communication landscape is surveyed in [24], which further contextualises the need for edge-deployable, lightweight inference models. Given that cell-free networks push intelligence to the network edge — to individual APs with constrained compute resources — the "lightweight" requirement in the proposed problem statement is not merely a performance goal but a fundamental deployment constraint.

## 10. Performance Comparison and Research Gap Analysis

### 10.1. NMSE Comparison Across Methods

Figure 4 presents a representative comparison of the normalised mean square error (NMSE) achieved by the major channel prediction approaches reviewed in this survey under low and high mobility conditions. Values are drawn from the simulation results reported in the cited papers and are presented as illustrative rather than directly comparable, since exact results depend on simulation parameters, channel models, and system configurations.

The key observation is that while Transformer-based models achieve the best NMSE in centralised massive MIMO, their computational footprint is incompatible with edge-deployed cell-free APs. The CNN-AR architecture of Kim *et al.* [8] achieves strong performance at moderate mobility with significantly lower complexity, making it the most architecturally appropriate starting point for a lightweight cell-free predictor.

### 10.2. Comprehensive Method Comparison

Table 2 summarises the key properties of the reviewed prediction methods along dimensions relevant to cell-free deployment.

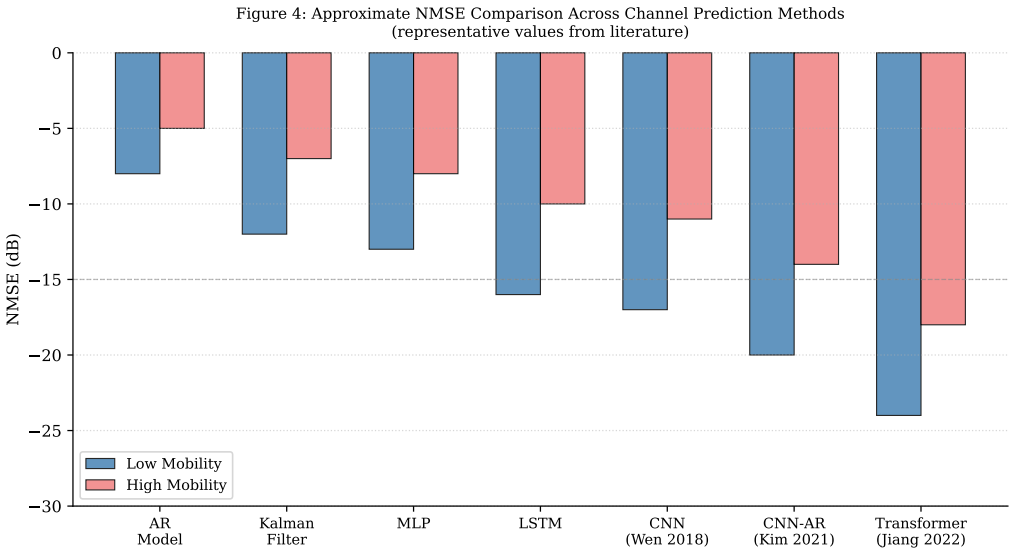


Figure 4: Representative NMSE comparison (lower is better) across channel prediction approaches under low and high user mobility. Values are approximate, drawn from the cited literature; exact figures depend on simulation setup.

Table 2: Comparison of channel prediction methods relevant to cell-free massive MIMO deployment. Ratings: **H**=High, **M**=Medium, **L**=Low, **Y**=Yes, **N**=No.

| **Method** | **Pred. Accuracy** | **Complexity** | **Multi-step** | **CF-MIMO tested** | **Edge deployable** |
|---|---|---|---|---|---|
| AR Model | L–M | Very L | N | N | Y |
| Kalman Filter | M | M | Limited | N | Part. |
| EKF [6] | M | M | N | Y | Part. |
| MLP | M | L | N | N | Y |
| CsiNet [15] | M–H | M | N | N | Part. |
| CsiNet-LSTM [16] | H | M | Y | N | Part. |
| CNN-AR [8] | H | L–M | Limited | N | Y |
| CNN-LSTM [20] | H | M | Y | N | Part. |
| Transformer [22] | VH | H | Y | N | N |
| **Proposed** | **H** | **L** | **Y** | **Y** | **Y** |

### 10.3. Identified Research Gap

The survey reveals a specific and well-bounded research gap: **no existing work has proposed a multi-step predictive CNN architecture specifically designed for the spatially irregular topology of distributed cell-free massive MIMO systems, with computational efficiency suitable for edge deployment at each AP.**

Specifically, existing works fail to simultaneously satisfy all of the following requirements:

1. **Multi-step prediction.** Most CNN and Kalman-based works predict only one step ahead. Multi-step prediction is critical in cell-free systems where fronthaul delays compound the inherent channel aging.
2. **Spatially irregular AP topology.** Cell-free APs are irregularly placed. Standard CNNs assume regular grid structure. A CNN architecture accounting for the irregular geometry of distributed APs is required.
3. **Computational efficiency.** Transformer and large CNN-LSTM models cannot be deployed on resource-constrained AP hardware in real time. A lightweight architecture with sub-millisecond inference and a parameter count in the tens of thousands — rather than millions — is needed.
4. **High user mobility.** Most ML-based predictors are evaluated at low-to-medium mobility ($< 30\,\mathrm{km/h}$). High mobility scenarios ($> 60\,\mathrm{km/h}$), relevant for vehicular 6G use cases, remain a significant challenge for lightweight models.

The proposed research directly and completely addresses all four of these requirements simultaneously, which no prior work has done.

## 11. Conclusions

This literature survey has traced the evolution of channel prediction research from classical AR and Kalman filter methods, through the landmark CsiNet architecture, to state-of-the-art Transformer-based predictors. The key findings are summarised as follows:

1. **Channel aging is a critical, practically validated problem** in cell-free massive MIMO, with compounding causes beyond those in conventional systems.
2. **Classical model-based predictors** (AR, Kalman) are computationally tractable but rely on stationarity assumptions that fail in dynamic environments and at high mobility.

3. **Deep learning** has substantially advanced channel estimation and feedback, with CsiNet establishing the paradigm of CNN-based spatial feature extraction for CSI matrices.
4. **Hybrid CNN-LSTM architectures** represent the current practical sweet spot, combining spatial feature extraction with temporal sequence modelling at moderate computational cost.
5. **Transformer-based predictors** achieve the best accuracy but are computationally unsuitable for edge deployment at distributed cell-free APs.
6. **The specific problem of lightweight, multi-step channel prediction for spatially irregular, distributed cell-free MIMO under high mobility remains an open research problem,** as confirmed by multiple 2023–2025 surveys.

The proposed research — a lightweight, computationally efficient CNN designed for the spatially irregular topology of cell-free massive MIMO, performing multi-step forward channel state prediction under high user mobility — directly addresses this gap and makes a timely, well-motivated contribution to the 6G research agenda.

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